# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 9 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 6: **How Forces Affect Motion** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** Assertion (A): A cricketer moves his hands backward while catching a fast ball. Reason (R): This increases the time of stopping and hence decreases the force on the hands. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**2.** A 60 kg man stands on a weighing machine in a lift. The reading is 72 kg when the lift moves. The lift's acceleration is (g = 10): 1

**(a)** 12 m/s squared upward **(b)** 2 m/s squared downward
**(c)** 2 m/s squared upward **(d)** zero

**3.** A book is placed on a table. The reaction to the weight of the book is: 1

**(a)** the normal force by the table **(b)** friction
**(c)** the push of the table **(d)** the pull of the book on the Earth

**4.** A 5 kg block on an incline of a smooth plane with a pull of 20 N along the incline (no other forces) accelerates at: 1

**(a)** 5 m/s squared **(b)** 20 m/s squared **(c)** 2 m/s squared **(d)** 4 m/s squared

**5.** Assertion (A): Action and reaction forces never cancel each other. Reason (R): They act on different bodies. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**6.** A 60 kg man stands on a weighing machine in a lift and reads 48 kg. The acceleration of the lift is: 1

**(a)** 8 m/s squared **(b)** 2 m/s squared upward
**(c)** zero **(d)** 2 m/s squared downward

**7.** A heavy box is pushed along the floor with a force F. Which of the following would make the push easier? 1

**(a)** increasing the weight **(b)** reducing the friction by rollers
**(c)** pushing downward **(d)** using a rough floor

**8.** A ball of mass 100 g is thrown with a speed of 20 m/s horizontally and hits the wall to stop in 0.01 s. The force is: 1

**(a)** 200 N **(b)** 2 N **(c)** 20 N **(d)** 2000 N

**9.** Assertion (A): A person standing in a lift feels heavier when the lift accelerates upward. Reason (R): The normal reaction from the floor is greater than his weight. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**10.** A balloon moving upward at constant velocity releases a stone. Just after release, the stone's acceleration is: 1

**(a)** zero **(b)** 20 m/s squared
**(c)** 10 m/s squared downward **(d)** 10 m/s squared upward

## SECTION — B (Very Short Answer Type) 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** A body of mass 2 kg moving with 10 m/s is acted on by a force of 4 N opposite to its motion. Find the time to stop and the distance. 2

**12.** Explain why a person at the surface of Earth does not move towards the Earth with acceleration g but is at rest. 2

**13.** Why does the tension in a rope used to pull a block at constant velocity equal the friction on the block? 2

**14.** Two forces of 9 N and 5 N act on a 2 kg body in the same direction and then in opposite directions. Find the accelerations. 2

## SECTION — C (Short Answer Type) 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** A car of mass 1500 kg is moving on a level road at 72 km/h. The driver applies the brakes and the car stops in 5 s. Find the force and the distance. Compare the force with the weight. 3

**16.** A bullet of mass 10 g moving at 400 m/s enters a wooden block and stops in 0.04 m. Find the average retarding force and the time of penetration. 3

**17.** A hammer of mass 1 kg moving at 10 m/s strikes a nail and stops in 0.01 s. Find the force and the force on the nail. 3

**OR**

A force of 30 N acts on a 6 kg body initially at rest for 4 s. Find the final velocity, the change in momentum and the distance covered. 3

**18.** A girl pushes a box with 40 N and the friction is 40 N. Describe the motion and explain what happens when she pushes with 60 N. 3

**OR**

Explain why the apparent weight of a person changes in an accelerating lift, giving three cases. 3

## SECTION — D (Long Answer Type) 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** A body of mass 8 kg starts from rest and is acted on by a net force of 24 N. Find the acceleration, the velocity and the distance after 5 s, and the momentum then. 5

**OR**

A cricket ball of mass 160 g approaches the bat at 30 m/s and leaves at 40 m/s in the opposite direction in 0.02 s. Find the impulse and the force, and explain how the force changes if the contact time is doubled. 5

**20.** A man of mass 70 kg stands on a weighing scale in a lift. Find the scale reading in kg when 5
(a) the lift is at rest
(b) it moves up at 3 m/s squared
(c) it moves down at 3 m/s squared
(d) the cable breaks (g = 10).

**OR**

An object of mass 5 kg is acted on by two forces: 30 N east and 40 N north. Find the magnitude of the net force, the acceleration and the direction. 5

## END OF QUESTION PAPER Total: 40 Marks